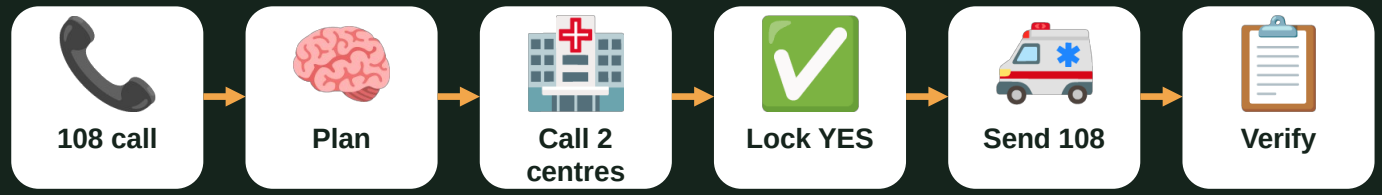
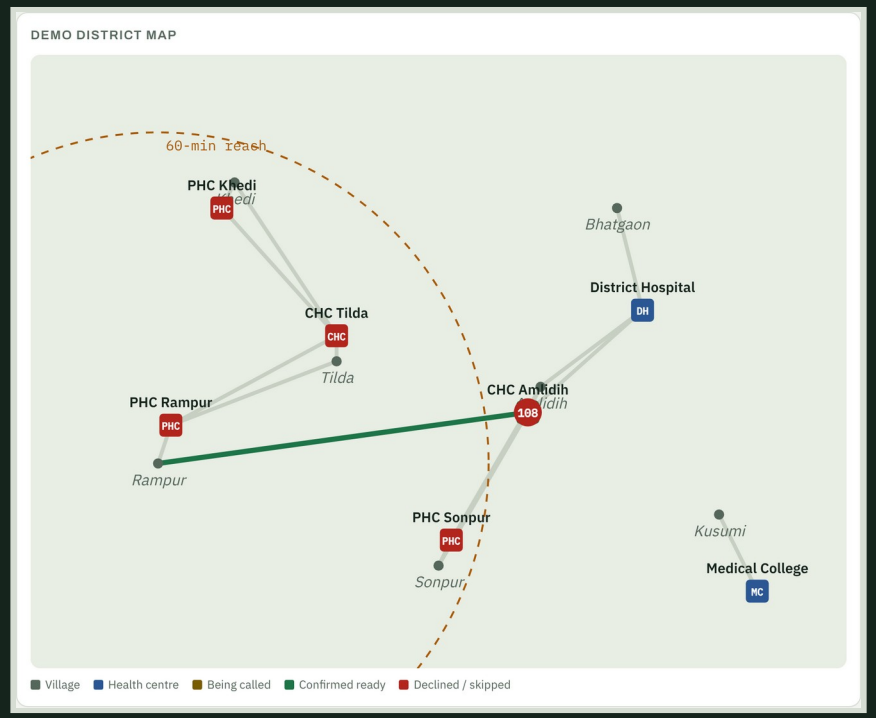
PROJECT ECLIPSE 2026 | ROUND 2 FINAL SOLUTION | OPEN-X TRACK

VishRaksha Agent

Snakebite to antivenom in 60 minutes.

An AI agent that reads the 108 call, finds centres that have antivenom and a doctor tonight, calls them by itself, locks the first YES, sends the ambulance there and checks the patient arrived. When something fails, it re-plans or calls a human.





Live demo: Tilda said no, Sonpur did not answer, so the agent re-planned and locked CHC Amlidih

TEAM INNOVERSE

Manas Girdhar Team Leader & Technical Lead, B.Tech, BIT Mesra
Moulvi Gufran Clinical partner, final-year MBBS

Live demo: https://claude.ai/artifact/GAnDQdzZFTtRXizbET7Wdn
Code: github.com/<your-id>/VishRaksha

PROBLEM

Ramu reached a centre in 15 min. Antivenom came 2 hours later.



23:20

Krait bite while sleeping on the floor, Rampur village



23:40

Family calls 108. Ambulance comes.



23:55

Nearest PHC: 0 vials of antivenom tonight



00:20

Referral slip, wait for another vehicle



01:20

Second centre: doctor not told, not ready



01:45

First antivenom. Paralysis already set in



~58,000

snakebite deaths a year in India, the highest in the world (Million Death Study)



Time

to antivenom decides survival. The killer is the wasted hour, not the distance



NAPSE

India's national plan aims to halve snakebite deaths by 2030

The gap: nobody checks, before the patient leaves, which centre has antivenom AND a doctor right now.

TARGET USERS AND EXISTING GAP

Everyone has a piece of the answer. Nobody joins them up at 11 PM.



108 control room

Moves the patient, but only to the nearest centre. Cannot see stock or doctor on duty.



PHC / CHC doctor

Gets the patient with no warning; often no antivenom or ventilator.



District health office

Finds out about stock-outs after a death, from paper registers.



Patient family

Panics, may go to a faith healer first. Will not open any app.

WHAT EXISTS TODAY, AND WHAT IS MISSING

	Knows stock?	Calls the centre?	Re-routes on NO?	Verifies arrival?
108 ambulance service	No	No	No	No
Drug stock systems (DVDMS / e-Aushadhi)	Partly, old data	No	No	No
Snake rescue apps (e.g. Kerala SARPA)	No	No	No	No
VishRaksha Agent	**Yes, daily + on call**	**Yes, by itself**	**Yes**	**Yes**

VishRaksha does not replace 108. 108 moves the patient; the agent picks and readies the centre.

PROPOSED SOLUTION

One agent core, three jobs: Reach, Rescue, Restock



REACH

Trigger: Every snakebite 108 call

- ✅ Reads the call note (Hinglish)
- ✅ Finds centres with antivenom + doctor within 60 min
- ✅ Calls the best 2 at once, press 1 / 2
- ✅ Locks the first YES, sends 108 there
- ✅ Briefs the centre, verifies arrival



RESCUE

Trigger: Doctor presses Refer

- ✅ Bedside timer: national protocol checks, doctor enters what they see
- ✅ On Refer: finds and confirms a ventilator bed
- ✅ Sends a handover note built from the doctor's entries
- ✅ Never reads a test, never suggests a drug



RESTOCK

Trigger: Every morning, 8 AM

- ✅ Asks each centre: how many vials today?
- ✅ No reply: voice call. Still none: mark unknown, tell officer
- ✅ Low stock: proposes a transfer
- ✅ A human approves before any vial moves

No app for the family. No app for the hospital. Only a phone that can take a call.

AI / AGENTIC AI APPROACH

The agent plans, uses tools, decides, acts and checks its own work

1. INPUT

108 note: "saanp ne kaata, Rampur, 20 min pehle"

2. UNDERSTAND

Claude reads Hinglish into JSON: snakebite, village, time. Unsure? Ask the caller

3. PLAN

Centres within 60 min with 10+ vials and a doctor; call best 2; widen if needed

4. TOOL USE

find_facilities, get_eta, ivr_call, send_whatsapp

5. DECISION

Skip PHC Rampur (0 vials). Tilda pressed 1 first: lock it

6. ACTION

notify_ambulance, pre-arrival note, stand-down to the other

7. VERIFY

At ETA: "Patient received? Press 1". Close and log

8. ESCALATE

Unclear place, all say no, stock moves: a human decides

Design rule: the LLM understands messy language; fixed, tested rules make the safety decisions (vial count, 60-min limit, fallback). So the agent is flexible with input and predictable with patients. If the LLM is down, a rule-based parser takes over.

Every step is time-stamped in a trace: the same trace judges see in the demo is the audit log.

RELIABILITY AND FAILURE HANDLING

Built for the bad night, not the good one



Centre does not answer in 2 min

TESTED

→ Next centre. The best 2 are always called together, so no time is lost



Centre presses 2 (not ready)

TESTED

→ Re-plan with the next centres on the list



No ready centre within 60 min

TESTED

→ Alert control room, widen to 90 min, keep going



Every centre declines

TESTED

→ Pre-agreed fallback (Medical College) + District officer alerted



Voice API returns an error

TESTED

→ Retry once, then WhatsApp with Ready / Not ready buttons



Village name unclear

TESTED

→ Confidence below 0.9: IVR asks caller to confirm; if no, a human takes over



Stock data over 24 h old

TESTED

→ Keep the centre but the call must confirm stock



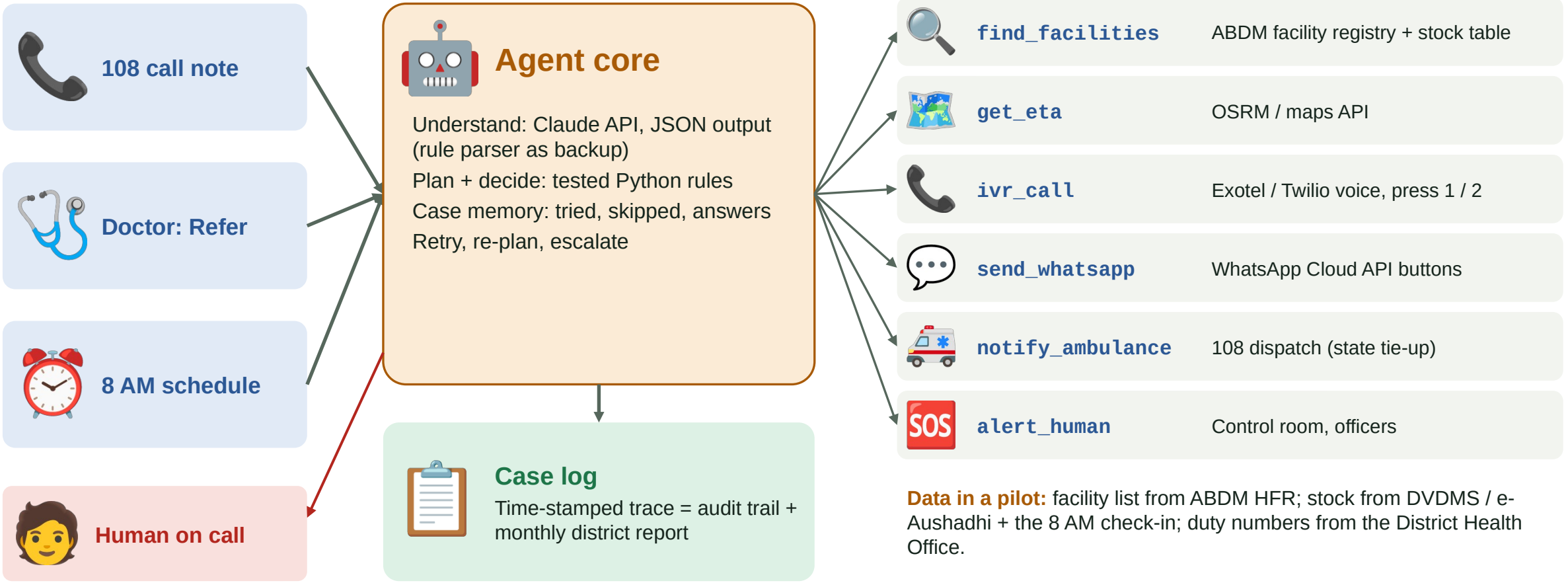
Not a snakebite

TESTED

→ Leave the case to the normal 108 flow

All 8 cases run in the demo or the 10 automated tests (pytest). A judge can make any centre fail live.

Three triggers, one agent core, real-world tools, a human on call



Data in a pilot: facility list from ABDM HFR; stock from DVDMS / e-Aushadhi + the 8 AM check-in; duty numbers from the District Health Office.

Hospitals need no new software: a voice call and a keypad are enough.

TECHNOLOGIES USED

Simple, cheap, India-ready stack

Claude API

Reads Hinglish 108 notes into structured JSON (village, time, snakebite yes/no)

Python 3.11 agent core

Typed tools, planner, retry and re-plan logic, case memory

pytest

10 automated tests: every failure path in the deck

Exotel / Twilio Voice

IVR calls in Hindi / English with keypad input (press 1 / 2)

WhatsApp Cloud API

Buttons for Ready / Not ready, stock check-in, nurse reminders

OSRM / Maps API

Road travel time from village to centre

ABDM HFR + DVDMS

Facility registry and government drug stock data

HTML + JavaScript demo

Runs the same agent logic offline in any browser

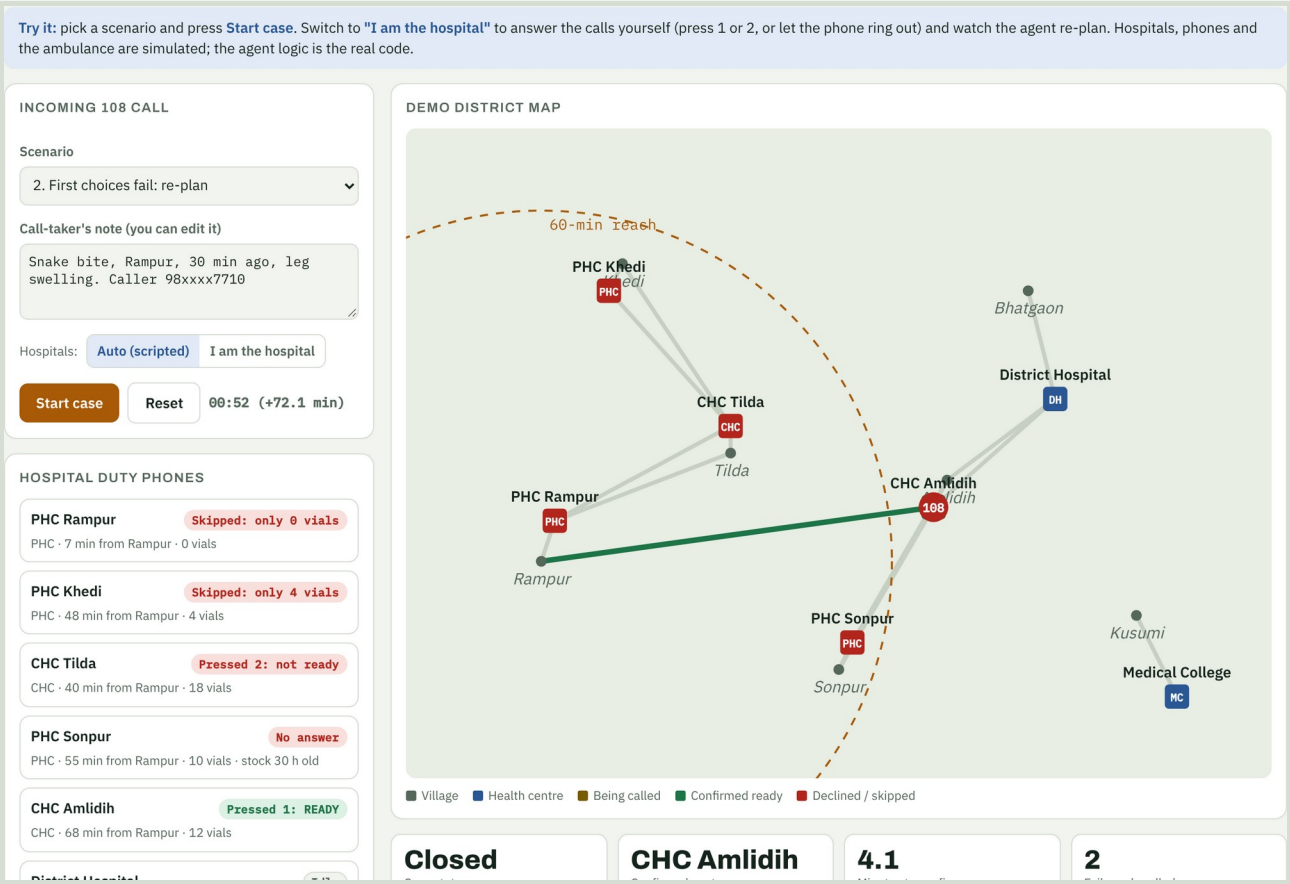
Python simulation

10,000-case monsoon model, fixed seed, editable assumptions

Running cost: about Rs 1.5-2 per call minute. Under Rs 50,000 per district for a whole monsoon.

CURRENT PROGRESS

Working today: try it at the demo link



Live case tab after the "first choices fail" scenario: phones, map, result tiles

BUILT

- ✅ Reach agent: 5 live scenarios
- ✅ Rescue agent: ventilator referral + handover note
- ✅ Restock agent: 8 AM check-in + approval step
- ✅ Claude call-note reader + rule backup
- ✅ 10 automated tests passing
- ✅ Simulation of 10,000 bites
- ✅ Public web demo + GitHub repo

NEXT (FOR THE FINALE)

- 🚀 Real calls to team phones via Exotel sandbox
- 🚀 WhatsApp test number for buttons
- 🚀 Hindi voice prompts

Honest scope: hospitals, phones and the ambulance are simulated; the agent logic is the real, tested code.

EXPECTED IMPACT

Simulation: 10,000 monsoon snakebites in one district

Reached a ready centre within 60 min



Ready centre at the first stop



Median minutes to a ready centre



Slowest 10% of patients, minutes



Model: 22 PHCs, 6 CHCs, 1 DH; 20 min lost per referral; 5% of YES answers wrong on arrival. Code: simulation/simulate.py. Planning numbers, not field data.

🎯 Pilot: one district, one monsoon

With the District Health Office, June to September. One metric: minutes from the 108 call to a centre that gives antivenom.

📦 Fewer stock-outs

The 8 AM agent finds low stock and empty replies every day, not after a death.

🔁 Same engine, more emergencies

Postpartum bleeding, dog bite (rabies vaccine), burns: any "who is ready right now?" problem.

From 64% to 92% of patients reaching a ready centre within the golden hour.

Safe by design, honest about limits



Safety controls

No diagnosis, no drug advice. The agent routes and reminds; doctors decide.

Ambulance never waits for the agent.

Humans approve high-impact steps.

Minimal data, masked numbers, logs deleted after closure (DPDP Act 2023).



Known limitations

Real duty numbers and the 108 link need state permission; the demo simulates them.

Stock is only as fresh as the 8 AM check-in, so each call re-confirms it.

Dialect speech can fail: keypad and a human operator are fallbacks.



Roadmap

Nov: real Exotel + WhatsApp numbers on team phones

Dec: clinical review of scripts with doctors

Monsoon 2027: one-district pilot

Later: monsoon antivenom demand forecast



VishRaksha Agent: snakebite to antivenom in 60 minutes.

Demo: https://claude.ai/artifact/GAnDQdzZFTtRXizbET7Wdn **Code:** github.com/<your-id>/VishRaksha

Thank you. Team InnoVerse